Some logical digital questions to think and practice

1. A PN flip flop has four operations, No change, clear to 0, set 1 and complement. When Inputs are P and N as 00,01,10 and 11
   a. Find characteristic table
   b. Derive characteristic equation
   c. Tabulate excitation table
   d. Show how PN FF can be converted to DFF
2. A combinational circuit is specified by the following Boolean functions

$$F_1(ABC) = \sum(3,5,6)$$
$$F_2(ABC) = \sum(1,4)$$
$$F_3(ABC) = \sum(2,3,5,6,7)$$

   Implement circuit with decoder. Minimize number of inputs in external gates

3. Implement using FIVE 4x1 mux (use A as input)

$$F(ABCDE) = \sum m(0,1,2,3,6,8,9,10,13,15,17,20,24,30)$$
$$[Hint: five\ 4\ x\ 1\ equivalent\ to\ 16\ x\ 1\ mux]$$

4. An engine has a 4 fail safe sensors. The engine should keep running unless any of the following conditions arise:

   If sensor 2 is activated

   If sensor 1 and sensor 3 are activated at same time

   If sensor 3 and sensor 4 are activated at same time

   Derive the truth table for the system (Hint: truth table for the system shut down)

   Design using k-map and give the implementation using NAND gates only, assume complemented inputs are available

5. An industry has uses two storage tanks to store some fluid that are required in a certain process. Each tank has a sensor that detects when the fluid level drops to 35% of full. The sensors produce a HIGH level of 5V when the tanks are more than three-quarter full. When the volume of fluid in a tank drops to three-quarter full, or less, the sensor puts out a LOW level of 0V. It is required that a single green light-emitting diode (LED) on an indicator panel shows when **both** tanks are **less** than one quarter full.
   a. Show the truth table
   b. Derive the Boolean expressions.
   c. Show how NAND gates can be used to implement this function
6. A certain system contains two identical circuits operating in parallel. As long as both are operating properly, the outputs of both circuits are always the same. If one of

the circuits fails, the outputs will be at opposite levels at some time. Devise a way to monitor and detect that a failure has occurred in one of the circuits. Also, find Will the exclusive-OR gate always detect simultaneous failures in both circuits of the above designed circuit?. If not, under what condition?

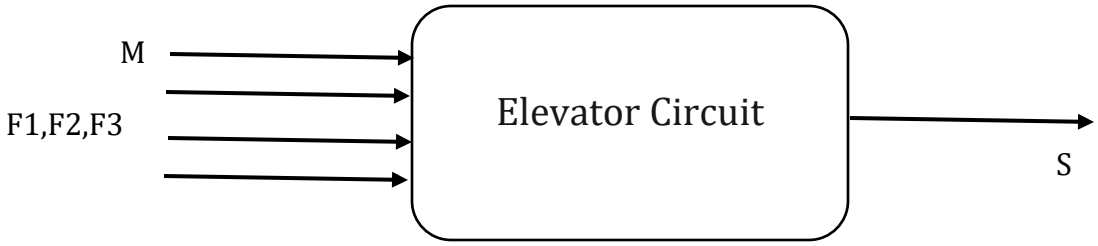
7. Design a combinational circuit with 4 input lines that represent a decimal digit in BCD and 4 output lines that generates the 9's compliment of the input digit
8. Design a combinational circuit which has input as a4 bit number and whose output is a 2's complement of the input number
9. Design a combinational circuit that multiplies by 2 an input decimal digit represented in BCD. The output is also in BCD
10. Design a combinational logic circuit that has 3 inputs A,B and C. its output will have the bit which would be in majority in the corresponding input combination.
11. A company has 4 directors A, B,C and D. the percentage of shares held by them are 35%, 30%, 25% and 10% respectively. These voting powers is proportional to shares held by them. Any major decision must have support of 60% or above of the stock. Design a combinational logic circuit for voting in the company.
12. A 4 bit binary number is represented as A3A2A1A0, where A0 is LSB and A3 is MSB. Design a logic circuit that will produce a high output whenever a binary number is greater than 0010 and less than 1000.
13. Design a multiplier circuit that takes 2 bit binary numbers X1 X0 and Y1 Y0. It produces an output binary number Z3 Z2 Z1 Z0 that is equal to the arithmetic product of two input numbers
14. Design vending machine for DH1 with three items capacity. Each item has a switch. The vending machine displays the cost and the Item number when the corresponding switch is pressed. The price list of the items is given below:

Item Number	cost (in rupees)
1	5
2	8
3	6
4	9

Design the logic between your Switch in the machine and the design using Seven Segment Display.

15. Assume a fuel tank is designed with the 10 sensors arranged on its wall at regular intervals. When the fuel is filled into the tank and the fuel touches each sensor the System you design has to calculate the fuel level and also display the level as 0,when empty and 9 when full at the last sensor. Design using multiplexer
16. Design a combinational circuit that controls an eleveator door in a three storey building in SNU. The circuit has four inputs as shown in figure below. One of the inputs M is a switch acting as a controller that indicates whether elevator is moving (M=1) or stopped (M=0). The other three inputs F1, F2 and F3 are floor indicator signals that are normally LOW and go HIGH only when elevator is positioned at the

level of that particular floor. For example when elevator is lined up at level with second floor F2=1 and F1=F3=0. The circuit output is S signal, which is normally LOW and will go HIGH when the elevator door is to be opened.



Design a circuit with minimum number of NAND gates (assume complemented inputs are available) if each NAND gate contributes to 8ns delay, calculate the circuit delay assuming all inputs are available at 0 ns

[Hint: Assume don't care wherever required]

17. Implement the following Boolean expressions with 3 half adders only (to be used as blocks)
    a. D= A xor B XOR C
    b. E=A'BC+AB'C
    c. F = ABC'+(A'+B')C
    d. G=ABC
18. FUNCTION F=f.g is defined as
    a. f=abc'+c'd+a'cd'+b'cd'
    b. g=(a+b+c'+d')(b'+c'+d)(a'+c+d')

    find the simplest SOP form of F

19. implement using 16:1 mux (use A as input)

    F(ABCDE) = ∑m(2,4,5,7,10,14,15,16,17,25,26,30,31)

20. combinational circuit is defined by the following expression
    F1 =a'b'+abc'
    F2 = a'+b
    F3 = ab+a'b'
    Design with a decoder and 3 external gates only. Gate 1 is a 3 input gate, gate 2 is a 4 input gate and gate 3 is a 2 input gate
21. design a 4 bit combinational circuit known as 2's complementer that generates 2's complement of input binary number. Design using XOR. Take inputs as ABCD and outputs as WXYZ.
22.
    a. Convert D to JK flipflop
    b. Construct a JK FF using a D FF, 2:1 mux and an inverter

[Hint: use expression derived in part a]

23. Realize full adder and full subtractor using 8x1 and 4x1 multiplexer
24. Implement a full subtractor using 3:8 decoder
25. Realize the following truth table using 4x1 mutiplexer

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

26. Realize the following operations using 2X1 mux. Assume two inputs A and B
    a. OR
    b. AND
    c. XOR
    d. XNOR
27. Design a GRAY to BCD converter using 1x16 demux
28.